$\therefore$ We can apply Cramer's Rule and the system is consistent and it has unique solution.

$$\Delta_x = \begin{vmatrix} 4 & 1 & 1 \\ 1 & -1 & 3 \\ 1 & 2 & -1 \end{vmatrix} = -13$$

$$\Delta_y = \begin{vmatrix} 1 & 4 & 1 \\ 2 & 1 & 3 \\ 3 & 1 & -1 \end{vmatrix} = 39 \quad \Delta_z = \begin{vmatrix} 1 & 1 & 4 \\ 2 & -1 & 1 \\ 3 & 2 & 1 \end{vmatrix} = 26$$

$\therefore$ By Cramer's rule

$$x = \frac{\Delta x}{\Delta} = \frac{-13}{13} = -1$$

$$y = \frac{\Delta y}{\Delta} = \frac{39}{13} = 3 \qquad z = \frac{\Delta z}{\Delta} = \frac{26}{13} = 2$$

$\therefore$ The solution is $(x, y, z) = (-1, 3, 2)$



If $|A|=0$, then the system of equations has either no solution or infinitely many solutions.

Example 1.23

The price of 3 Business Mathematics books, 2 Accountancy books and one Commerce book is ₹ 840. The price of 2 Business Mathematics books, one Accountancy book and one Commerce book is ₹ 570. The price of one Business Mathematics book, one Accountancy book and 2 Commerce books is ₹ 630. Find the cost of each book by using Cramer's rule.

Solution:

Let 'x' be the cost of a Business Mathematics book

Let 'y' be the cost of a Accountancy book.

Let 'z' be the cost of a Commerce book.

$$\therefore \quad 3x + 2y + z = 840$$
$$2x + y + z = 570$$
$$x + y + 2z = 630$$

Here

$$\Delta = \begin{vmatrix} 3 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 2 \end{vmatrix} = -2 \neq 0$$

$$\Delta_x = \begin{vmatrix} 840 & 2 & 1 \\ 570 & 1 & 1 \\ 630 & 1 & 2 \end{vmatrix} = -240$$

$$\Delta_y = \begin{vmatrix} 3 & 840 & 1 \\ 2 & 570 & 1 \\ 1 & 630 & 2 \end{vmatrix} = -300$$

$$\Delta_z = \begin{vmatrix} 3 & 2 & 840 \\ 2 & 1 & 570 \\ 1 & 1 & 630 \end{vmatrix} = -360$$

$\therefore$ By Cramer's rule

$$x = \frac{\Delta x}{\Delta} = \frac{-240}{-2} = 120$$

$$y = \frac{\Delta y}{\Delta} = \frac{-300}{-2} = 150 \qquad z = \frac{\Delta z}{\Delta} = \frac{-360}{-2} = 180$$

$\therefore$ The cost of a Business Mathematics book is ₹120, the cost of a Accountancy book is ₹150 and the cost of a Commerce book is ₹180.

Example 1.24

An automobile company uses three types of Steel S_1, S_2 and S_3 for providing three different types of Cars C_1, C_2 and C_3. Steel requirement R (in tonnes) for each type of car and total available steel of all the three types are summarized in the following table.

Types of Steel	Types of Car			Total Steel available
	C_1	C_2	C_3	
S_1	3	2	4	28
S_2	1	1	2	13
S_3	2	2	1	14

Determine the number of Cars of each type which can be produced by Cramer's rule.

Solution:

Let 'x' be the number of cars of type C_1

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